2.1 Set Theory

2.1.1 Relations

Definition 2.1.1 (Binary Relation)
*A **binary relation** (or just **relation**) R on a pair of sets (X, Y) is subset of the cartesian product $X \times Y$.*

We write xRy to mean precisely $(x, y) \in R$.

Definition 2.1.2 (Converse Relation)
Let R be a binary relation on (X, Y) then we the converse relation R^T on (Y, X) given by

$$yR^T x \iff xRy$$

Definition 2.1.3 (Domain and Range)
*Let R be a relation on (X, Y). We define the **domain** of R to be*

$$\text{dom}(R) \coloneqq \{x \in X \mid \exists y \in Y \text{ s.t. } xRy\}$$

*and the **range** of R*

$$\text{range}(R) \coloneqq \{y \in X \mid \exists x \in X \text{ s.t. } xRy\}$$

Definition 2.1.4 (Equivalence Relation)
Let R be a binary relation on (X, X). It is said to be

*a) **reflexive** if xRx for all $x \in X$*
*b) **symmetric** if $xRy \implies yRx$ for all $x, y \in X$*
*c) **transitive** if $xRy \wedge yRz \implies xRz$ for all $x, y, z \in X$*

*A relation which satisfies all these properties is called an **equivalence relation** on X. In this case we would write*

$$x \sim y$$

*instead of xRy. For an element $x \in X$ denote the **equivalence class** of x by*

$$[x]_R = \{y \mid xRy\}$$

Note that $R^T = R$.

Definition 2.1.5 (Partition)
*Let X be a set an $\mathcal{F}$ a family of subsets of X. It is said to be a **partition** if*

a) $X = \bigcup_{A \in \mathcal{F}} A$
b) $A, B \in \mathcal{F} \implies A = B$ or $A \cap B = \emptyset$

Proposition 2.1.6 (Equivalence Classes form a Partition)
Let E be an equivalence relation on X. The family

$$\mathcal{F} = \{[x]_E \mid x \in X\}$$

*forms a partition of X. Denote by X/E the family of equivalence classes, called the **quotient** of X with respect to E.*

Proof. It's clear that

$$X = \bigcup_{A \in \mathcal{F}} A$$

because by reflexive-ness $x \in [x]_E$ for all $x \in X$.

We claim that for any $z \in [x]_E$ we have $[z]_E = [x]_E$. Suppose $y \in [x]_E$ then xRz and xRy. By symmetry and transitivity we then have zRy which implies $y \in [z]_E$. In other words $[x]_E \subseteq [z]_E$. By symmetry of R we have $x \in [z]_E$, so by the same token $[z]_E \subseteq [x]_E$, which shows they are equal.

Therefore it's clear that $[x]_E \cap [y]_E \neq \emptyset \implies [x]_E = [y]_E$ and thus $\mathcal{F}$ forms a partition. ∎

Definition 2.1.7 (Composition of Relation)
Suppose R is a relation on (X, Y) and S a relation on (Y, Z). We define the composition $S \circ R$ on (X, Z)

$$S \circ R = \{(x, z) \mid \exists y \in Y \text{ s.t. } xRy \text{ and } yRz\}$$

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2.1.2 Functions

Definition 2.1.8 (Function)
*A **function** $f : X \to Y$ consists of a binary relation $\Gamma(f)$ on (X, Y) such that*

- $\mathrm{dom}(f) = X$
- $\Gamma(f)$ *is* ***single-valued*** *that is* $x\Gamma(f)y \wedge x\Gamma(f)y' \implies y = y'$

Equivalently for all $x \in X$ there exists precisely one $y \in Y$ such that $x\Gamma(f)y$.

We write $f(x) = y$ for the unique element $y \in Y$ such that $x\Gamma(f)y$.

Proposition 2.1.9 (Equality of Functions)
Two functions $f, g : X \to Y$ are equal if and only if $f(x) = g(x)$ for all $x \in X$.

Proposition 2.1.10 (Composition of Functions)
Let $f : X \to Y$ and $g : Y \to Z$ be functions then the composition $\Gamma(g) \circ \Gamma(f)$ is still a function, which we write $g \circ f$, and

$$(g \circ f)(x) = g(f(x))$$

*Furthermore composition is **associative** in the sense that*

$$(h \circ g) \circ f = h \circ (g \circ f)$$

Definition 2.1.11 (Injective, Surjective and Bijective)
Let $f : X \to Y$ be a function then we say

- *f is **injective** if $f(x) = f(x') \implies x = x'$*
- *f is **surjective** if for all $y \in Y$ there exists x such that $f(x) = y$*
- *f is **bijective** if it is both injective and surjective*

Definition 2.1.12 (Inverse Function)
Let $f : X \to Y$ and $g : Y \to X$ be functions. We say

- *g is a **left inverse** for f if $g \circ f = 1_X$*
- *g is a **right inverse** for f if $f \circ g = 1_Y$*
- *g is a **two-sided inverse** for f if it is both a left and right inverse*

Proposition 2.1.13
Let $f : X \to Y$ be a function then

- *f is **injective** if and only if it has a **left inverse***
- *f is **surjective** if and only if it has a **right inverse***
- *f is **bijective** if and only if it has a **two-sided inverse***

Definition 2.1.14 (Idempotent Function)
*A function $p : X \to X$ is **idempotent** $p \circ p = p$.*

Lemma 2.1.15 (Idempotent Criterion)
Let $p : X \to X$ be a function. Then $\mathrm{Fix}(p) \subseteq \mathrm{Im}(p)$ and these are equal if and only if p is idempotent.

2.1.3 Partial Orders

Definition 2.1.16 (Preorder)
*A binary relation $\leq$ on (X, X) is a **preorder** if*

- ***reflexivity** $x \leq x$*
- ***transitivity** $x \leq y$ and $y \leq z \implies x \leq z$*

*We may refer to $(X, \leq)$ as a **preordered set** or **preorder**.*

Definition 2.1.17 (Poset)
*A binary relation $\leq$ on (X, X) is a **partial order** if*

- ***reflexivity** $x \leq x$*

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- ***antisymmetry*** $x \leq y$ and $y \leq x \implies y = x$
- ***transitivity*** $x \leq y$ and $y \leq z \implies x \leq z$

*We may refer to $(X, \leq)$ as a **partially ordered set** or **poset**. Equivalently, a poset is a preorder which is additionally antisymmetric.*

Definition 2.1.18 (Dual Poset)
*Given a poset $(X, \leq)$ denote the set X with the converse relation by $\left(X, \overset{d}{\leq}\right)$. This is the **dual poset** to $(X, \leq)$.*

Example 2.1.19
Let $\mathcal{F}$ be a family of subsets of a fixed set E. Then $(\mathcal{F}, \subseteq)$ is a poset ordered under inclusion.

Definition 2.1.20 (Top and Bottom)
*Let $(X, \leq)$ we say $\top$ (resp. $\bot$) is a **top element** (resp. **bottom element**) if it is greater than (resp. less than) every element of x. In this case it is unique.*

Definition 2.1.21 (Monotone/Antitone Function)
Let $(X, \leq)$ and $(Y, \leq)$ be posets. A function $f : X \to Y$ is

- ***monotone / order-preserving*** *if $x \leq y \implies f(x) \leq f(y)$*
- ***antitone / order-reversing*** *if $x \leq y \implies f(y) \leq f(x)$*
- *a **monotone embedding** if $x \leq y \iff f(x) \leq f(y)$*
- *an **order isomorphism** if it is bijective and monotone*
- *a **dual isomorphism** if it is bijective and antitone*

Proposition 2.1.22
Let $f : X \to Y$ be a monotone function. Then it is an embedding if and only if it is injective.

In what follows the notion of closure and kernel operator will be important.

Definition 2.1.23 (Closure operator)
*Let $(X, \leq)$ be a partially ordered set. A function $c : X \to X$ is a **closure operator** if it is*

a) ***extensive*** $x \leq c(x)$
b) ***monotone*** $x \leq y \implies c(x) \leq c(y)$
c) ***idempotent*** $c(c(x)) = c(x)$

Definition 2.1.24 (Kernel operator)
*Let $(X, \leq)$ be a partially ordered set. A function $\kappa : X \to X$ is a **kernel operator** if it is*

- ***co-extensive*** $\kappa(x) \leq x$
- ***monotone*** $x \leq y \implies \kappa(x) \leq \kappa(y)$
- ***idempotent*** $\kappa(\kappa(x)) = \kappa(x)$

Note these definitions are "dual" with respect to the ordering on X.

2.1.4 Lattices

Certain families of subsets of algebraic structures (e.g. ideals, subgroups, normal subgroups, submodules) form a "sublattice" of the power set. Certain operations on, and results about, these subsets share common features regardless of the type of algebraic structure. Therefore we detail some elements of "Lattice Theory" (see Birkhoff) which may clarify the exposition.

Definition 2.1.25 (Upper and Lower Bounds)
*Let $(X, \leq)$ be a poset and $S \subseteq X$. Define the set of **upper bounds** for S by*

$$S^{\uparrow} = \{x \in X \mid s \leq x \quad \forall s \in S\}$$

*and the set of **lower bounds** for S by*

$$S^{\downarrow} = \{x \in X \mid x \leq s \quad \forall s \in S\}$$

Note by convention $\emptyset^{\uparrow} = \emptyset^{\downarrow} = X$. Furthermore

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$$X^{\uparrow} = \begin{cases} \{\top\} & X \text{ has a top element} \\ \emptyset & \text{otherwise} \end{cases}$$

and

$$X^{\downarrow} = \begin{cases} \{\bot\} & X \text{ has a bottom element} \\ \emptyset & \text{otherwise} \end{cases}$$

Lemma 2.1.26 (Upper/Lower bounds are antitone maps)
Let $(X, \leq)$ be a poset and S, T subsets of X then

- ***antitone*** $S \subseteq T \implies T^{\uparrow} \subseteq S^{\uparrow}$ *and* $T^{\downarrow} \subseteq S^{\downarrow}$
- ***unit-counit relations*** $S \subseteq S^{\uparrow\downarrow}$ *and* $T \subseteq T^{\downarrow\uparrow}$
- ***triangular identities*** $S^{\uparrow} = S^{\uparrow\downarrow\uparrow}$ *and* $T^{\downarrow} = T^{\downarrow\uparrow\downarrow}$

Proof. We prove only the first triangular identity as the others are straightforward consequences of the definitions. Firstly $S \subseteq S^{\uparrow\downarrow} \implies S^{\uparrow\downarrow\uparrow} \subseteq S^{\uparrow}$ by the antitone property. Given the relation $T \subseteq T^{\downarrow\uparrow}$ substitute $T = S^{\uparrow}$ to get the reverse inclusion. ■

Lemma 2.1.27
*Let $(X, \leq)$ be a poset and S, T subsets of X. Then the intersections $S \cap S^{\uparrow}$ and $T \cap T^{\downarrow}$ contain at most one element. **When they exist** write the elements as $\top_S$ and $\bot_T$ respectively, and are referred to as the **maximum** and **minimum** elements respectively.*

Proof. Given $x, y \in S \cap S^{\uparrow}$ then by definition $x \leq y$ and $y \leq x$. By anti-symmetry we have $x = y$ as required. ■

Definition 2.1.28 (Supremum and Infimum)
*Let $(X, \leq)$ be a poset and $S \subseteq X$ a subset. We say a **supremum** of S is the minimal upper bound, i.e. the unique element of*

$$S^{\uparrow} \cap S^{\uparrow\downarrow}$$

*when it exists and write this as $\sup S$. Similarly an **infimum** of S is the maximal lower bound, i.e. the unique element of*

$$S^{\downarrow} \cap S^{\downarrow\uparrow}$$

when it exists and write this as $\inf X$.

Lemma 2.1.29 (Maximum = Supremum)
Let $(X, \leq)$ be a poset and $S \subseteq X$ a subset. Then $\top_S$ exists if and only if $\sup S$ exists and is a member of S. In this case $\top_S = \sup S$.

Lemma 2.1.30
Let $(X, \leq)$ be a poset. Then $\{\sup S\}^{\uparrow} = S^{\uparrow}$ and $\{\inf T\}^{\downarrow} = T^{\downarrow}$ when these exist.

Lemma 2.1.31 (Sup is monotone and Inf is antitone)
Let $(X, \leq)$ be a poset and S, T subsets of X. Then $S \subseteq T \implies \sup S \leq \sup T$ and $\inf T \leq \inf S$ when these exist.

Proof. Note $S \subseteq T \implies T^{\uparrow} \subseteq S^{\uparrow}$ so $\sup T \in S^{\uparrow}$. By definition $\sup S \in S^{\uparrow\downarrow}$ therefore $\sup S \leq \sup T$.

Similarly $S \subseteq T \implies T^{\downarrow} \subseteq S^{\downarrow}$. By definition $\inf T \in T^{\downarrow} \implies \inf T \in S^{\downarrow}$. By definition $\inf S \in S^{\downarrow\uparrow}$ therefore $\inf T \leq \inf S$.
■

Remark 2.1.32
Note that $\emptyset^{\uparrow} = X$ and therefore $\sup \emptyset = \bot$ when it exists. Similarly $\inf \emptyset = \top$ when it exists.

When $\top$ exists $\sup X = \top$, otherwise it is not defined. Similarly when $\bot$ exists $\inf X = \bot$, otherwise it is not defined.

Definition 2.1.33 (Lattice)
*A poset $(X, \leq)$ is a **lattice** if every pair of elements $\{x, y\}$ admits both a supremum and infimum. In this case we write*

$$a \vee b := \sup\{a, b\}$$

and

$$a \wedge b := \inf\{a, b\}$$

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*These are called the **join** and **meet** operations. A subset Y is called a **sub-lattice** if*

$$a, b \in Y \implies a \wedge b \in Y \text{ and } a \vee b \in Y.$$

*Similarly it is a **complete lattice** if every subset S admits both a supremum and infimum. This is written*

$$\bigvee S := \sup S$$

and

$$\bigwedge S := \inf S$$

*Note a complete lattice has both a top and a bottom element (by considering $\sup \emptyset$ and $\inf \emptyset$), and a lattice admits **finite** joins and meets.*

Trivially

$$\bigwedge \{x\} = \bigvee \{x\} = x$$

Example 2.1.34 (Power Set)
*For a fixed set E the collection of subsets $\mathcal{P}(E)$ is a complete lattice under the union and intersection operator **with the convention that empty intersection is the whole set and empty union is the empty set***

In this case $\top = E$ and $\bot = \emptyset$.

Proposition 2.1.35 (Principal down-sets are lattices)
Let $(X, \leq)$ be a lattice and $x, y \in X$. Then the subsets $\{x\}^\uparrow$, $\{x\}^\downarrow$ and $\{x\}^\uparrow \cap \{y\}^\downarrow$ are sub-lattices.

Verifying a poset is a lattice is slightly easier than it may first appear.

Lemma 2.1.36 (Supremum is Infimum of upper bounds)
Let $(X, \leq)$ be a poset and S a subset of X. Then

$$\sup S = \inf S^\uparrow$$

when either exists. Dually

$$\inf S = \sup S^\downarrow$$

Proof. By definition $\sup S$ is the unique element of $S^\uparrow \cap S^{\uparrow\downarrow}$ and $\inf S^\uparrow$ is the unique element of $S^{\uparrow\downarrow} \cap S^{\uparrow\downarrow\uparrow}$. By (2.1.26) $S^{\uparrow\downarrow\uparrow} = S^\uparrow$ so they are equivalent. ∎

Proposition 2.1.37 (Criteria to be a Complete Lattice)
Let $(X, \leq)$ be a poset. Then the following are equivalent

a) X is a complete lattice
b) X admits arbitrary infimums (and in particular has $\top = \inf \emptyset$)
c) X admits arbitrary supremums (and in particular has $\bot = \sup \emptyset$)

In this case we have the relationships

$$\bigvee S = \bigwedge S^\uparrow$$

$$\bigwedge S = \bigvee S^\downarrow$$

Proof. a) $\implies$ b,c) is clear.

b,c) $\implies$ a) follows from the previous Lemma. ∎

Lemma 2.1.38
Let $(X, \leq)$ be a poset and $(Y, \leq)$ a sub-poset. Let $S \subseteq Y$ be a subset. Then $\inf_Y S$ exists if and only if $\inf_X S$ exists and belongs to Y. In this case they are equal.

Proof. Note in general that $T^{\downarrow,Y} = T^{\downarrow,X} \cap Y$ and $T^{\uparrow,Y} = T^{\uparrow,X} \cap Y$. Therefore

$$S^{\downarrow,Y} \cap S^{\downarrow\uparrow,Y} = S^{\downarrow,X} \cap S^{\downarrow\uparrow,X} \cap Y$$

Recall $\inf S$ is the unique element of $S^\downarrow \cap S^{\downarrow\uparrow}$ if it exists. Then the result follows easily. ∎

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Definition 2.1.39 (Moore Family)
*Let $(X, \leq)$ be a complete lattice. A sub-poset $(Y, \leq)$ is a **Moore family** over X if it satisfies the following property*

$$S \subseteq Y \implies \bigwedge_X S \in Y$$

In particular this includes the case $S = \emptyset$ and so $\top \in Y$.

Example 2.1.40 (Moore family of sets)
Given a fixed set E, then $\mathcal{P}(E)$ is a complete lattice ordered under inclusion. Then a family of subsets $\mathcal{F}$ is a Moore family precisely when

- $E \in \mathcal{F}$
- $A_{i \in I} \in \mathcal{F} \implies \bigcap_{i \in I} A_i \in \mathcal{F}$

Proposition 2.1.41 (Equivalent Formulations of Complete Sub-lattice)
Let $(X, \leq)$ be a complete lattice and $(Y, \leq)$ a sub-poset. Then the following are equivalent

a) $(Y, \leq)$ is a Moore family
b) $(Y, \leq)$ is a complete lattice
c) Y is the image of some closure operator $c : X \to X$

In this case the closure operator is given by

$$c(x) = \bigwedge_X \{y \in Y \mid x \leq y\}$$

For $S \subseteq Y$

$$\bigwedge_Y S = \bigwedge_X S$$

$$\bigvee_Y S = c\left(\bigvee_X S\right)$$

and for $S \subseteq X$ we have

$$c\left(\bigvee_X S\right) = \bigwedge_X (S^\uparrow \cap Y)$$

Proof. a) $\implies$ b) By (2.1.38) $S \subseteq Y \implies \bigwedge_Y S = \bigwedge_X S$. By (2.1.37) then Y is a complete lattice.

b) $\implies$ c) Suppose that $(Y, \leq)$ is a complete lattice then define the function $c : X \to X$ by $c(x) = \bigwedge_X \Gamma_x$ where $\Gamma_x = \{y \in Y \mid x \leq y\}$. We need to show that it is a closure operator. Evidently $x \in \Gamma_x^\downarrow$ and $c(x) \in \Gamma_x^{\downarrow\uparrow}$ by definition of infimum. Therefore $x \leq c(x)$ and c is extensive. Note $x \leq y \implies \Gamma_y \subseteq \Gamma_x$. By (2.1.31) we have $\inf \Gamma_x \leq \inf \Gamma_y$, whence $c(x) \leq c(y)$ and c is monotone.

Y is a complete lattice, so by (2.1.38) we have $c(x) \in Y$ so that $\mathrm{Im}(c) \subseteq Y$. We claim that $x \in Y \implies c(x) = x$. In this case $x \in \Gamma_x$ and $c(x) \in \Gamma_x^\downarrow$ whence $c(x) \leq x$ and therefore $x = c(x)$ as required. Therefore $Y \subseteq \mathrm{Fix}(c) \subseteq \mathrm{Im}(c) \subseteq Y$, whence $Y = \mathrm{Im}(c) = \mathrm{Fix}(c)$ and c is idempotent by (2.1.15). As c is extensive, monotone and idempotent it is by definition a closure operator.

c) $\implies$ a) In order for $Y \coloneqq \mathrm{Im}(c)$ to be a Moore family, we need to show $S \subseteq Y \implies \bigwedge_X S \in Y$. We claim that by properties of c we have

$$S \subseteq Y \implies c\left(S^\downarrow\right) \subseteq S^\downarrow$$

$$T \subseteq X \implies c(T^\uparrow) \subseteq T^\uparrow$$

Therefore c maps the singleton set $S^\downarrow \cap S^{\downarrow\uparrow} = \left\{\bigwedge_X S\right\}$ to itself. In otherwords $\bigwedge_X S \in \mathrm{Fix}(c) = \mathrm{Im}(c) = Y$ as required.

Define $\Gamma_x \coloneqq \{y \in Y \mid x \leq y\}$. We wish to show that $c(x) = \bigwedge_X \Gamma_x$. As $x \leq c(x)$ we have $c(x) \in \Gamma_x$. Furthermore $y \in \Gamma_x \implies c(x) \leq c(y) = y$. So $c(x) \in \Gamma_x^\downarrow$. Therefore $c(x) = \bot_{\Gamma_x} = \bigwedge_X \Gamma_x$ as required.

Finally by (2.1.30) $\left\{\bigvee_X S\right\}^\uparrow = S^\uparrow$ for any $S \subseteq X$. Therefore, as $c(x) = \bigwedge_X \Gamma_x$ we find

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$$c\left(\bigvee_X S\right) = \bigwedge_X \left(\left\{\bigvee_X S\right\}^{\uparrow} \cap Y\right) = \bigwedge_X (S^{\uparrow} \cap Y)$$

as required. In particular when $S \subseteq Y$ we find by (2.1.41)

$$\bigvee_Y S = \bigwedge_Y S^{\uparrow, Y} = \bigwedge_X S^{\uparrow, Y} = \bigwedge_X (S^{\uparrow} \cap Y) = c\left(\bigvee_X S\right)$$

■

Remark 2.1.42
For a given complete lattice $(X, \leq)$ we have established a correspondence between

$$\{\text{closure operators } \ c : X \to X\} \longleftrightarrow \{\text{complete sub-lattices } \ (Y, \leq)\}$$

Corollary 2.1.43 (Moore family admits a closure operator)
Let E be a fixed set and $\mathcal{F}$ a Moore family over $(\mathcal{P}(E), \subseteq)$. Then there exists a surjective closure operator $c : \mathcal{P}(E) \to \mathcal{F}$ given by

$$c(F) = \bigcap_{F \subseteq E_\alpha \in \mathcal{F}} E_\alpha$$

Any such closure operator $c : \mathcal{P}(E) \to \mathcal{P}(E)$ gives rise to a Moore family $\mathcal{F} := \mathrm{Im}(c)$.

Proposition 2.1.44 (Alternative expression for join)
Let $(X, \leq)$ be a complete lattice and $c : X \to X$ a closure operator with image Y. Then for any subset $S \subset X$

$$c\left(\bigvee_X S\right) = c\left(\bigvee_X c(S)\right) = \bigvee_Y c(S) = \bigwedge_X (S^{\uparrow} \cap Y)$$

i.e. it's the smallest "closed" set containing each element of S.

Proof. By (2.1.41) the expression for $c\left(\bigvee_X S\right)$ yields

$$c\left(\bigvee_X c(S)\right) = \bigwedge_X (c(S)^{\uparrow} \cap Y) = \bigwedge_X (S^{\uparrow} \cap Y) = c\left(\bigvee_X S\right)$$

where the middle equality follows because if $y \in Y$ then $c(s) \leq y \iff s \leq y$. Furthermore $c(S) \subseteq Y$ so the expression for $\bigvee_Y$ yields

$$c\left(\bigvee_X c(S)\right) = \bigvee_Y c(S).$$

as required. ■

2.1.5 Distributive Lattice

Proposition 2.1.45
Let $(X, \leq)$ be a lattice then the following relations hold

a) $x \wedge (y \vee z) \geq (x \wedge y) \vee (x \wedge z)$
b) $x \vee (y \wedge z) \leq (x \vee y) \wedge (x \vee z)$

Proof. a) By definition $x \wedge y \leq x$ and $x \wedge y \leq y \leq y \vee z$. Therefore $x \wedge y \leq x \wedge (y \vee z)$. By symmetry in y and z we have $x \wedge z \leq x \wedge (y \vee z)$. Whence $(x \wedge y) \vee (x \wedge z) \leq x \wedge (y \vee z)$ as required.

b) follows by duality ■

Definition 2.1.46 (Distributive Lattice)
*We say a lattice $(X, \leq)$ is **distributive** if it satisfies the following relations for all $x, y, z \in X$*

- $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$
- $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$

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Proposition 2.1.47
Let $(X, \leq)$ be a lattice then TFAE

a) $(X, \leq)$ *is a distributive lattice*
b) $x \wedge (y \vee z) \leq (x \wedge y) \vee (x \wedge z)$
c) $x \vee (y \wedge z) \geq (x \vee y) \wedge (x \vee z)$

Example 2.1.48
Any family of subsets closed under finite intersection and union is a distributive lattice.

2.1.6 Galois Connections

Definition 2.1.49 (Galois Connection)
Let $\left(X, \underset{X}{\leq}\right)$ and $\left(Y, \underset{Y}{\leq}\right)$ be posets. A pair of functions $(f_\star, f^\star)$

$$X \underset{f^\star}{\overset{f_\star}{\rightleftharpoons}} Y$$

*is called an **antitone Galois connection** if it satisfies the **adjoint property***

- $x \underset{X}{\leq} f^\star(y) \iff y \underset{Y}{\leq} f_\star(x) \quad \forall x \in X, y \in Y$

*We say it is a **monotone Galois connection** if instead*

- $x \underset{X}{\leq} f^\star(y) \iff f_\star(x) \underset{Y}{\leq} y \quad \forall x \in X, y \in Y$

We will assume that if not otherwise specified the connection is antitone.

Proposition 2.1.50 (Equivalent Condition for Galois Connection)
Let $\left(X, \underset{X}{\leq}\right)$ and $\left(Y, \underset{Y}{\leq}\right)$ be posets. Consider a pair of functions

$$X \underset{f^\star}{\overset{f_\star}{\rightleftharpoons}} Y$$

*Then this constitutes an **antitone Galois Connection** if and only if*

- $f_\star$ *and* $f^\star$ *are both antitone*
- $x \underset{X}{\leq} f^\star(f_\star(x))$ *and* $y \underset{Y}{\leq} f_\star(f^\star(y))$ *for all* $x \in X, y \in Y$ *(i.e.* $f^\star \circ f_\star$ *and* $f_\star \circ f^\star$ *are extensive)*

*Similarly it constitutes a **monotone Galois Connection** if and only if*

- $f_\star$ *and* $f^\star$ *are both monotone*
- $x \underset{X}{\leq} f^\star(f_\star(x))$ *and* $f_\star(f^\star(y)) \underset{Y}{\leq} y$ *for all* $x \in X, y \in Y$

Proof. We consider only the antitone case, as the monotone follows from duality (flip $\underset{Y}{\leq}$).

Suppose that $(f_\star, f^\star)$ satisfies the adjoint property

$$f_\star(x) = f_\star(x) \implies f_\star(x) \underset{Y}{\leq} f_\star(x) \implies x \underset{X}{\leq} f^\star(f_\star(x))$$

$$f^\star(y) = f^\star(y) \implies f^\star(y) \underset{Y}{\leq} f^\star(y) \implies y \underset{Y}{\leq} f_\star(f^\star(y))$$

whence the extensive property follows. Furthermore

$$x \underset{X}{\leq} x' \implies x \underset{X}{\leq} f^\star(f_\star(x')) \implies f_\star(x') \underset{Y}{\leq} f_\star(x)$$

$$y \underset{Y}{\leq} y' \implies y \underset{Y}{\leq} f_\star(f^\star(y')) \implies f^\star(y) \underset{X}{\leq} f^\star(y')$$

which shows that the functions $f_\star$ and $f^\star$ are antitone.

Conversely suppose they satisfy the given conditions. Then by the antitone and extensive properties in turn

$$x \underset{X}{\leq} f^\star(y) \implies f_\star(f^\star(y)) \underset{Y}{\leq} f_\star(x) \implies y \underset{Y}{\leq} f_\star(x)$$

and

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$$y \underset{Y}{\leq} f_\star(x) \implies f^\star(f_\star(x)) \underset{X}{\leq} f^\star(y) \implies x \underset{X}{\leq} f^\star(y).$$

which is the adjoint property as required. ∎

Definition 2.1.51 (Closed sets)
*Let $(f_\star, f^\star)$ be a Galois connection. Then define the **closed sets** to be*

$$X^\star := f^\star(Y)$$

$$Y^\star := f_\star(X)$$

Proposition 2.1.52 (Isomorphism on closed sets)
*Consider an antitone (resp. monotone) Galois connection $X \underset{f^\star}{\overset{f_\star}{\rightleftarrows}} Y$. Then it restricts to a dual isomorphism (resp. order isomorphism) on **closed sets***

$$X^\star \underset{f^\star}{\overset{f_\star}{\rightleftarrows}} Y^\star$$

Furthermore the following properties hold

- $f_\star \circ f^\star \circ f_\star = f_\star$
- $f^\star \circ f_\star \circ f^\star = f^\star$
- *In the antitone case $f^\star \circ f_\star$ and $f_\star \circ f^\star$ are closure operators.*
- *In the monotone case $f^\star \circ f_\star$ is a closure operator and $f_\star \circ f^\star$ is a kernel operator.*
- $X^\star = \mathrm{Fix}(f^\star \circ f_\star) = \mathrm{Im}(f^\star \circ f_\star)$
- $Y^\star = \mathrm{Fix}(f_\star \circ f^\star) = \mathrm{Im}(f_\star \circ f^\star)$

Proof. We detail the antitone case as the monotone case follows by duality. We first prove so-called triangular identities, for by the extensive property (2.1.50)

$$x \leq f^\star(f_\star(x)) \implies f_\star(f^\star(f_\star(x))) \leq f_\star(x)$$

and by the other extensive property

$$f_\star(x) \leq f_\star(f^\star(f_\star(x)))$$

whence they are equal. The other case is similar.

It's immediate that $f^\star \circ f_\star$ and $f_\star \circ f^\star$ are idempotent, and they are extensive by (2.1.50). And the composition of two antitone functions is monotone so $f^\star \circ f_\star$ and $f_\star \circ f^\star$ are closure operators.

Observe

$$\mathrm{Im}(f^\star \circ f_\star) \subseteq \mathrm{Im}(f^\star) \subseteq \mathrm{Fix}(f^\star \circ f_\star)$$

where the first inclusion is trivial and the second inclusion follows from the second triangular identity. However both sides are equal by (2.1.15) and the expression for $X^\star$ follows. The expression for $Y^\star$ follows similarly.

This shows that the maps are mutual inverses as required. ∎

In certain circumstances we may consider a smaller subset of X, by applying a suitable closure operator which is compatible with the Galois correspondence:

Proposition 2.1.53 (Subordinated Closure Operator)
Let $X \underset{f^\star}{\overset{f_\star}{\rightleftarrows}} Y$ be a Galois connection and $c : X \to X$ be a closure operator with image X_c. Then

$$c(x) \leq (f^\star \circ f_\star)(x) \quad \forall x \in X \iff \mathrm{Im}(f^\star) \subseteq \mathrm{Fix}(c) \iff X^\star \subseteq X_c$$

In this case

$$f_\star(c(x)) = f_\star(x)$$

Proof. Suppose $c(x) \leq (f^\star \circ f_\star)(x)$. Substitute $x = f^\star(y)$ then, because c is extensive,

$$f^\star(y) \leq c(f^\star(y)) \leq (f^\star \circ f_\star \circ f^\star)(y) = f^\star(y).$$

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Therefore $c(f^\star(y)) = f^\star(y)$ and $\mathrm{Im}(f^\star) \subseteq \mathrm{Fix}(c)$ as required. Conversely suppose this holds, then by the monotone property of c and extensive property of $f^\star \circ f_\star$

$$x \leq (f^\star \circ f_\star)(x) \implies c(x) \leq c((f^\star \circ f_\star)(x)) = (f^\star \circ f_\star)(x).$$

as required. Finally by the extensive property of c

$$x \leq c(x) \leq (f^\star \circ f_\star)(x)$$

and by the antitone/monotone property of $f_\star$ and triangular identity

$$f_\star(x) \leq f_\star(c(x)) \leq f_\star(x)$$

whence $f_\star(c(x)) = f(x)$ as required. ∎

The meaning of the "adjoint" criterion can be explained by the following rather generic situation

Example 2.1.54 (Canonical example of an antitone Galois connection)
Suppose there is a predicate

$$\psi : X \times Y \to \{0, 1\}$$

Define a connection

$$\mathcal{P}(X) \underset{f^\star}{\overset{f_\star}{\rightleftarrows}} \mathcal{P}(Y)$$

by

$$f_\star(S) = \{y \in Y \mid \psi(x, y) = 1 \quad \forall x \in S\}$$

$$f^\star(T) = \{x \in X \mid \psi(x, y) = 1 \quad \forall y \in T\}$$

Then

$$S \subseteq f^\star(T) \iff \psi(s, t) = 1 \quad \forall s \in S \quad t \in T \iff T \subseteq f_\star(S)$$

Proposition 2.1.55 (Joins under Galois Correspondence)
Let $\left(X, \underset{X}{\leq}\right)$ *and* $\left(Y, \underset{Y}{\leq}\right)$ *be complete lattices with an antitone Galois connection* $X \underset{f^\star}{\overset{f_\star}{\rightleftarrows}} Y$*. Then for* $S \subseteq X$

$$f_\star\Big(\bigvee S\Big) = \bigwedge f_\star(S)$$

Similarly for $T \subseteq Y$ *we have*

$$f^\star\Big(\bigvee T\Big) = \bigwedge f^\star(T)$$

Proof. Let $a = \bigvee S$ and $b = \bigwedge f_\star(S)$. Then $s \leq a \implies f_\star(a) \leq f_\star(s)$ for all $s \in S$, which implies $f_\star(a) \leq b$. Similarly $b \leq f_\star(s) \implies s \leq f^\star(b)$ by the adjoint criterion. Therefore $a \leq f^\star(b)$ by definition of join, which implies $b \leq f_\star(a)$ by the adjoint criterion again. Whence $f_\star(a) = b$ as required.

The second statement follows from duality. ∎

2.1.7 Axiom of Choice

Theorem 2.1.56 (Axiom of choice)
There are a number of essentially equivalent forrmulations of the axiom of choice

a) The Cartesian product of a non-empty family of sets is non-empty
b) For any set X of non-empty sets there exists a function $f : X \to \bigcup X$ such that $A \in X \implies f(A) \in A$.
*c) **Zorn's Lemma** Suppose a partially ordered set $(X, \leq)$ is such that every chain in X has an upper bound in X. Then X contains at least one maximal element.*
d) Every surjective function has a right inverse.

Corollary 2.1.57 (Choose representatives)
Let $\pi : X \to Y$ be a surjective function and $T \subseteq Y$ a subset. Then there exists a subset $S \subseteq X$ such that $\pi|_S$ is bijective.

When T is finite $\#S = \#T$.

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Definition 2.1.58 (Finite Character)
*A family of sets $\mathcal{F}$ has **finite character** if it satisfies the following property*

$$A \in \mathcal{F} \iff (B \subseteq A \text{ and finite} \implies B \in \mathcal{F})$$

Corollary 2.1.59 (Tukey's Lemma)
Let $\mathcal{F}$ be a family of sets of finite character. Then it is chain-complete when ordered by inclusion.

In particular every set is contained in a maximal set.

2.1.8 Chain Conditions

Definition 2.1.60 (Totally ordered / chains)
*A poset $(\mathcal{F}, \leq)$ is **totally ordered** if $x \leq y$ or $y \leq x$ for all $x, y \in \mathcal{F}$.*

Definition 2.1.61 (Chain)
*A non-empty subset C of $\mathcal{F}$ is a **chain** if it is totally ordered under $\leq$.*

The length of the chain is simply $\ell(C) := |C| - 1$.

A chain C is

- ***saturated** if $x \leq z \leq y$ and $x, y \in C \implies z \in C$.*
- ***maximal** if it's not contained properly in another chain.*

Definition 2.1.62 (Chain-Complete)
*A poset $(\mathcal{F}, \leq)$ is **chain complete** if every chain C has a supremum in $\mathcal{F}$. It is **co-chain complete** if every chain C has an infimum in $\mathcal{F}$.*

Proposition 2.1.63 (Noetherian / Artinian Poset)
Let $(X, \leq)$ be a poset then the following conditions are equivalent

a) Any ascending chain

$$x_1 \leq x_2 \leq ... \leq x_n \leq ...$$

eventually stabilizes

b) Any non-empty subset $Y \subseteq X$ has a maximal element

*Such a poset is called **Noetherian**. If it satisfies the dual condition then it is called **Artinian**.*

Proof. a) $\implies$ b) If Y has no maximal elements then we may (by axiom of dependent choice) construct a strictly increasing sequence, which by definition does not stabilize.

b) $\implies$ a) Clear. ■

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